Bodhitha (Bo) Jayatilaka

Fermi National Accelerator Laboratory · M.S. 205 · P.O. Box 500 · Batavia, IL 60510 USA
Tel: +1 630 840 5110 · boj@fnal.gov · bodhitha.github.io · ORCID: 0000-0001-7912-5612

Education

University of Michigan, Ann Arbor, Michigan USA
Ph.D., Physics 2006
Dissertation: *"A Measurement of the Top Quark Mass in the Dilepton Decay Channel at CDF II"*
Advisor: Prof. David Gerdes
M.S., Physics 2005

University of California, Berkeley, California USA
B.A., Physics 2002

University of Chicago Booth School of Business, Chicago, Illinois USA
Strategic Laboratory Leadership Program (Non-Degree) 2022–2023

Appointments

Fermi National Accelerator Laboratory (Fermilab), Batavia, Illinois USA
Division Leader 2026–present
Senior Scientist 2023–present
Scientist 2019–2023
Applications Physicist II 2016–2019
Applications Physicist I 2012–2016

Massachusetts Institute of Technology (MIT), Cambridge, Massachusetts USA
Visiting Scientist, Laboratory for Nuclear Science 2014–2018

Duke University, Durham, North Carolina USA
Research Scientist 2011–2012
Postdoctoral Research Associate 2006–2011

Fermilab Management Roles

Physics Directorate
Division Leader, Energy Frontier 2026–present

Particle Physics Directorate
Director, Collider Physics Division 2024–2026

Computational Science and AI Directorate
Deputy Director, Scientific Computing Systems and Services Division 2022–2024
Department Head, Facility Evolution 2024

LHC Physics Center
Co-Coordinator 2021–2025

Scientific Computing Division
Interim Department Head, Data Movement and Storage 2019
Group Leader, Storage Services Administration 2017–2019
Deputy Department Head, Data Movement and Storage 2016–2019
Technical Lead, CDF Tevatron Run 2 Data Preservation Project 2012–2014

Collaboration Leadership Roles

US Higgs Factory Circular Collider Steering Group
L1 Co-Coordinator, Software and Computing 2026–present

Open Science Grid

Area Coordinator for Production Support	2015–2016
Production Analyst	2014–2016

CDF Collaboration

Standard Model Physics Co-Convener	2013–2022
Operations Manager (Run Coordinator)	2011
W Boson Mass and Width Working Group Co-Convener	2008–2012
Calibrations Coordinator	2007–2008

Scientific Collaborations

CMS Experiment	2014–present
Fermilab Team Leader (2024–present)	
Dark Energy Survey	2013–2014
ATLAS Experiment	2011–2012
CDF Experiment	2003–present

Grants

DOE FAIR — *"Preparing for the Exascale of the CMS experiment."*
Mar 2025–Mar 2027. $250k. Co-I (PI: Dr. Allison Hall, US Naval Academy).

Fermilab Research B&R (KA21) — Energy Frontier Research
Sep 2024–present. ∼$14M annually. PI.

US CMS Software and Computing Research Initiative — *"Object Storage for CMS in the HL-LHC era."*
Jan 2022–Jan 2024. $195k. PI.

Honors and Awards

Fermilab Exceptional Performance Recognition Award	2026
Fermilab Reward and Recognition Award	2023
Fermilab Exceptional Performance Recognition Award	2021
Fermilab Reward and Recognition Award	2020
Fermilab Reward and Recognition Award	2018
Fermilab Reward and Recognition Award	2016
URA Alvin Tollestrup Award for Postdoctoral Research	2012

Service

Community

US LHC Users Association Executive Committee	2025–2027
DPF Graduate Instrumentation Research Award Selection Committee	2022
HEPiX Forum Board Member and Institutional Representative	2020–2023
External Reviewer, Holland Computing Center, University of Nebraska	2019
Frontiers in Big Data, Machine Learning in HEP; Associate Editor	2019–present
HEP Software Foundation Community White Paper Editorial Board Member	2017–2018

Fermilab

Scientific Long Range Planning Committee	2025
Committee on Scientific Appointments Policy	2025–2026
Wilson Fellowship Selection Committee	2021–2024
CMS Department Research Associate Mentoring Committee	2021–present
Media Policy Task Force, Chair	2020–2021
Scientist Advisory Council, Co-Chair / Past Chair	2019–2020 / 2020–2021
CMS Department Research Associate Hiring Committee	2019–2021
Asian/Pacific Islander-American Association, Co-Leader	2018–2021

CMS Collaboration

Diversity Monitoring and Oversight Committee	2024–2025
US CMS Elections Committee	2024–2026
LHC Physics Center Management Board, Co-Chair / Member	2021–2025 / 2025–present
Collaboration Board Advisory Group	2021–2025
Task Force on Diversity and Inclusion	2020–2021
CMS Diversity Office, Co-Chair / Member	2018–2020 / 2020–2023

Supervision

Dr. Murtaza Safdari, Postdoc	2022–present
Dr. Oz Amram, Postdoc; *subsequently Wilson Fellow at Fermilab, now industry AI researcher*	2022–2026
Dr. T. Christian Herwig, Postdoc; *now faculty at University of Michigan*	2021–2024
Dr. Karri DiPetrillo, Lederman Fellow/Postdoc; *now faculty at University of Chicago*	2021–2022
Dr. Nicholas Smith, Postdoc; *now staff at Fermilab*	2018–2024
Dr. Allison Reinsvold Hall, Postdoc; *now faculty at US Naval Academy*	2018–2021
Weinan Si (UC Riverside), LPC Graduate Scholar	2018–2020
Dr. Matteo Cremonesi, Postdoc; *now faculty at Carnegie Mellon University*	2015–2021

Selected Publications

I am an author in over 1,300 publications, primarily with the CMS and CDF collaborations, listed at inspire-hep.net/authors/1021024. Selected publications with major personal contribution are listed below.

2026 The CMS Collaboration (A. Hayrapetyan, V. Makarenko, B. Jayatilaka et al.), "Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV," Accepted for publication by Eur. Phys. J. C [arXiv:2604.20999].

2024 The CMS Collaboration (A. Hayrapetyan, A. Tumasyan, B. Jayatilaka et al.), "Search for Soft Unclustered Energy Patterns in Proton-Proton Collisions at 13 TeV," *Phys. Rev. Lett.* 133, 191902 [arXiv:2403.05311].

2024 C. Herwig, J. Isaacson, B. Jayatilaka et al., "Shedding light on the MiniBoone Excess with Searches at the LHC," *Phys. Rev. D* 109, 075049 [arXiv:2310.13042].

2024 The CMS Collaboration (A. Hayrapetyan, A. Tumasyan, B. Jayatilaka et al.), "Search for inelastic dark matter in events with two displaced muons and missing transverse momentum at $\sqrt{s} = 13$ TeV," *Phys. Rev. Lett.* 132, 041802 [arXiv:2305.11649].

2023 T. Basaglia, B. Jayatilaka et al. (DPHEP Collaboration), "Data Preservation in High Energy Physics — DPHEP Global Report 2022," *Eur. Phys. J. C* 83, 795 [arXiv:2302.03583].

2023 W. Bhimji, B. Jayatilaka et al., "Snowmass 2021 CompF4 Topical Group Report: Storage and Processing Resource Access," *Comput. Softw. Big Sci.* 7, 5 [arXiv:2209.08868].

2022 T. Aaltonen, B. Jayatilaka et al. (CDF Collaboration), "High-precision measurement of the W boson mass with the CDF II detector," *Science* 376, 170.

2021 S. Sekmen, G. M. Innocenti, B. Jayatilaka, "Editorial: Innovative Analysis Ecosystems for HEP Data," *Front. Big Data* 4:736105.

2021 The CMS Collaboration (A. Tumasyan, W. Adam, B. Jayatilaka et al.), "Search for new particles in events with energetic jets and large missing transverse momentum at $\sqrt{s} = 13$ TeV," *J. High Energ. Phys.* 2021, 153 [arXiv:2107.13021].

2020 The CMS Collaboration (Sirunyan, A.M., Tumasyan, A., Jayatilaka B. et al.), "Search for dark matter particles produced in association with a Higgs boson at $\sqrt{s} = 13$ TeV," *J. High Energ. Phys.* 2020, 25 [arXiv:1908.01713].

2019 The CMS Collaboration (Sirunyan, A.M., Tumasyan, A., Jayatilaka B. et al.), "Search for dark matter produced in association with a Higgs boson decaying to $b\bar{b}$ at $\sqrt{s} = 13$ TeV," *Eur. Phys. J. C* 79, 280 [arXiv:1811.06562].

2019 The HEP Software Foundation (Albrecht J., Jayatilaka B. et al.), "A Roadmap for HEP Software and Computing R&D for the 2020s," *Comput. Softw. Big Sci.* 3, 7 [arXiv:1712.06982].

2018 The CMS Collaboration (Sirunyan, A.M., Tumasyan, A., Jayatilaka B. et al.), "Search for dark matter in events with hadronically decaying top quarks and missing transverse momentum at $\sqrt{s} = 13$ TeV," *J. High Energ. Phys.* 2018, 27 [arXiv:1801.08427].

2017 The CMS Collaboration (Sirunyan, A.M., Tumasyan, A., Jayatilaka B. et al.), "Search for dark matter produced with an energetic jet or W/Z boson at $\sqrt{s} = 13$ TeV," *J. High Energ. Phys.* 2017, 14 [arXiv:1703.01651].

2017 S. Amerio, B. Jayatilaka et al., "Data preservation at the Fermilab Tevatron," *Nucl. Instrum. Methods Phys. Res. Sect. A* 851, 1 [arXiv:1701.07773].

2016 A. V. Kotwal and B. Jayatilaka, "Comparison of HORACE and PHOTOS Algorithms for Multiphoton Emission in the Context of W Boson Mass Measurement," *Adv. High Energy Phys.* 2016, 1615081 [arXiv:1510.02458].

2014 T. Aaltonen, B. Jayatilaka et al. (CDF Collaboration), "A precise measurement of the W-boson mass with the CDF II Detector," *Phys. Rev. D* 89, 072003 [arXiv:1311.0894].

2013 T. Aaltonen, B. Jayatilaka et al. (CDF & DØ Collaborations), "Combination of CDF and DØ W-boson mass measurements," *Phys. Rev. D* 88, 052108 [arXiv:1307.7627].

2012 T. Aaltonen, B. Jayatilaka et al. (CDF & DØ Collaborations), "Evidence for a Particle Produced in Association with Weak Bosons and Decaying to $b\bar{b}$ in Higgs Boson Searches at the Tevatron," *Phys. Rev. Lett.* 109, 071804. **PRL Editors' Suggestion** [arXiv:1207.6436].

2012 T. Aaltonen, B. Jayatilaka, *et al.* (CDF Collaboration), "Search for the Standard Model Higgs Boson Decaying to a $b\bar{b}$ Pair in Events with Two Oppositely Charged Leptons Using the Full CDF Data Set," Phys. Rev. Lett. 109, 11803 (2012). [arXiv:1207.1704].

2012 T. Aaltonen, B. Jayatilaka, *et al.* (CDF Collaboration), "Search for the Standard Model Higgs Boson Produced in Association with a Z boson in 7.9 fb^{-1} of $p\bar{p}$ Collisions at $\sqrt{s} = 1.96$ TeV Using the CDF II Detector," Phys. Lett. B 715, 98 (2012).

2012 T. Aaltonen, B. Jayatilaka et al. (CDF Collaboration), "Precise Measurement of the W Boson Mass with the CDF II Detector," *Phys. Rev. Lett.* 108, 151803. **PRL Editors' Suggestion** [arXiv:1203.0275].

2012 T. Aaltonen, B. Jayatilaka et al. (CDF Collaboration), "Combined Search for the Standard Model Higgs Boson Decaying to $b\bar{b}$ Using the Full CDF Data Set," *Phys. Rev. Lett.* 109, 111802 [arXiv:1207.1707].

2012 T. Aaltonen, B. Jayatilaka, *et al.* (CDF Collaboration), "Search for New Phenomena in Events with Two Z Bosons and Missing Transverse Momentum in $p\bar{p}$ Collisions at $\sqrt{s} = 1.96$ TeV," Phys. Rev. D 85, 011104(R) (2012) [arXiv:1112.1577].

2010 T. Aaltonen, B. Jayatilaka et al. (CDF Collaboration), "Top Quark Mass Measurement in the Lepton + Jets Channel Using a Matrix Element Method and *in situ* Jet Energy Calibration," *Phys. Rev. Lett.* 105, 252001 [arXiv:1010.4582].

2010 T. Aaltonen, B. Jayatilaka, *et al.* (CDF Collaboration), "Improved Search for a Higgs Boson Produced in Association with $Z \to l^+l^-$ in $p\bar{p}$ Collisions at $\sqrt{s} = 1.96$ TeV," Phys. Rev. Lett. 105, 251802 (2010). [arXiv:1009.3047].

2009 T. Aaltonen, B. Jayatilaka, *et al.* (CDF Collaboration), "A Search for the Higgs Boson Produced in Association with $Z \to \ell^+\ell^-$ Using the Matrix Element Method at CDF II," Phys. Rev. D 80, 071101(R) (2009). [arXiv:0908.3534].

2009 T. Aaltonen, B. Jayatilaka et al. (CDF Collaboration), "Top Quark Mass Measurement in the $t\bar{t}$ All Hadronic Channel Using a Matrix Element Technique," *Phys. Rev. D* 79, 072010 [arXiv:0811.1062].

2009 T. Aaltonen, B. Jayatilaka, *et al.* (CDF Collaboration), "Measurement of the Top-Quark Mass with Dilepton Events Selected Using Neuroevolution at CDF," Phys. Rev. Lett. 102, 152001 (2009). [arXiv:0807.4652].

2008 T. Aaltonen, B. Jayatilaka et al. (CDF Collaboration), "First Run II Measurement of the W Boson Mass at the Fermilab Tevatron," *Phys. Rev. D* 77, 112001 [arXiv:0708.3642].

2007 T. Aaltonen, B. Jayatilaka et al. (CDF Collaboration), "First Measurement of the W Boson Mass in Run II of the Tevatron," *Phys. Rev. Lett.* 99, 151801 [arXiv:0707.0085].

2007 A. Abulencia, B. Jayatilaka et al. (CDF Collaboration), "Precision Measurement of the Top-Quark Mass from Dilepton Events at CDF II," *Phys. Rev. D* 75, 031105(R) [hep-ex/0612060].

2006 A. Abulencia, B. Jayatilaka, *et al.* (CDF Collaboration), "Top-Quark Mass Measurement from Dilepton Events at CDF II," Phys. Rev. Lett. 96, 152002 (2006). [hep-ex/0512070].

2006 A. Abulencia, B. Jayatilaka, *et al.* (CDF Collaboration), "Top-Quark Mass Measurement from Dilepton Events at CDF II Using the Matrix-Element Method," Phys. Rev. D. 74, 032009 (2006). [hep-ex/0605118].

2006 J. Antos, B. Jayatilaka et al., "Data Production Models for the CDF Experiment," *IEEE Trans. Nucl. Sci.* 53, 2897 [hep-ex/0606042].

Selected Talks and Presentations

2023 "Electroweak Results from the Tevatron and HERA," CERN Symposium: Electroweak Milestones, Geneva, Switzerland, October 2023. *Invited talk.*

2022–23 "Precision measurement of the W boson mass at CDF" — Invited seminars at University of Minnesota, Princeton University, Johns Hopkins/University of Maryland, University of Michigan, University of Wisconsin–Madison, Karlsruhe Institute of Technology, University of Chicago.

2022 "Software and Computing for the LHC and HL-LHC," 55th Annual Fermilab Users Meeting, Batavia, IL. *Invited plenary talk, on behalf of CMS.*

2020 "Fermilab Data Storage Experience and Planning," HSF-WLCG Virtual Workshop.

2020 "Diversity and Inclusion in the CMS Collaboration," ICHEP 2020, 40th International Conference on High Energy Physics.

2016 "Taking HEP data management outside of HEP" CHEP 2016, San Francisco, CA.

2016 "The OSG Open Facility," CHEP 2016, San Francisco, CA.

2016 "Searches for Dark Matter Production with CMS," LHCP 2016, Lund, Sweden. *Invited talk, on behalf of CMS.*

2015 "Data preservation at the Fermilab Tevatron," DPF 2015, Ann Arbor, MI.

2015 "Data preservation at the Fermilab Tevatron," CHEP 2015, Okinawa, Japan.

2015 "The OSG open facility: A sharing ecosystem," CHEP 2015, Okinawa, Japan.

2014 "M_W, Z A_{FB}, and W/Z Production," LHCP 2014, New York, NY. *Invited plenary talk, on behalf of ATLAS, CMS, CDF, and DØ.*

2013 "Legacy electroweak results from the Tevatron," Invited seminar, University of Cincinnati.

2012–13 "Precision measurement of the W boson mass at CDF" — Invited seminars at Wayne State University, Harvard University, MIT, University of Warwick, University of Birmingham, LAPP Annecy, Michigan State University, University of Notre Dame, University of California– Davis, Northwestern University, SLAC National Accelerator Laboratory, University of California– Los Angeles, University of California–Irvine, Argonne National Laboratory.

2012 "Recent results from the Tevatron," 2nd MCTP Symposium on Higgs Boson Physics, Ann Arbor, MI. *Invited plenary talk, on behalf of CDF and DØ.*

2012 "Measurement of the W boson mass at CDF," Rencontres de Moriond: Electroweak, La Thuile, Italy. *Invited plenary talk, on behalf of CDF.*

2012 "Measurement of the W boson mass at CDF," Les Rencontres de Physique de la Vallée d'Aoste, La Thuile, Italy. *Invited plenary talk, on behalf of CDF.*

2011 "Combination of SM Higgs Boson searches at the Tevatron," Rencontres de Moriond: Electroweak, La Thuile, Italy. *Invited plenary talk, on behalf of CDF and DØ.*

2010 "Search for the Low Mass SM Higgs Boson at the Tevatron," Aspen Particle Physics 2010, Aspen, CO. *Invited plenary talk, on behalf of CDF and DØ.*

2010 "Searching for the Standard Model Higgs Boson at CDF" — Invited seminars at University of California, Irvine and Columbia University.

2009 "Search for $VH \to \not\!\!E_T + b\bar{b}$ at the Tevatron," DPF 2009. Detroit, Michigan USA. *Invited talk, on behalf of the CDF and DØ.*

2009 "Tevatron Electroweak Results and Top Quark Properties," 42nd Annual Fermilab Users' Meeting, Batavia, IL. *Invited plenary talk, on behalf of CDF and DØ.*

2009 "Precision Electroweak Measurements at the Tevatron," DIS 2009, Madrid, Spain. *Invited talk, on behalf of CDF and DØ.*

2008 "Top Quark Mass Measurements in the Dilepton and Alljets Channels," ICHEP 2008, Philadelphia, PA. *Invited talk, on behalf of CDF and DØ.*

2006 "Top Mass Measurement with Matrix Element Technique using Dilepton Channel at CDF," DPF+JPS 2006, Honolulu, HI

2006 "Precision Measurements of the Top Quark Mass in the Dilepton Channel," Top 2006, Coimbra, Portugal. *Invited plenary talk, on behalf of CDF and DØ.*

Conference and Meeting Organization

(C) Conference, (M) Meeting, (S) School, (W) Workshop

2026 (C) APS Division of Particles and Fields (DPF26), Batavia, IL. *Local Organizing Committee Member.*

2025 (M) US Future Circular Collider Annual Meeting, Batavia & Lemont, IL. *Local Organizing Committee Member.*

2025 (C) APS Conference for Undergraduate Women and Gender Minorities in Physics (CU*iP), Batavia, IL. *Local Organizing Committee Member.*

2025 (S) XXXI CMS Data Analysis School, Batavia, IL. *Organizing Committee Co-Chair.*

2024 (S) XXIX CMS Data Analysis School, Batavia, IL. *Organizing Committee Co-Chair.*

2023 (S) XXVI CMS Data Analysis School, Batavia, IL. *Organizing Committee Co-Chair.*

2022 (S) XXV CMS Data Analysis School, Virtual. *Organizing Committee Co-Chair.*

2020 (M) US CMS Annual Collaboration Meeting, Virtual. *Program Committee Member.*

2020 (C) Snowmass Community Planning Meeting, Virtual. *Local Organizing Committee Co-Chair; Program Committee Ex Officio Member.*

2020 (W) 3rd Rucio Community Workshop, Batavia, IL. *Local Organizing Committee Member.*

2019 (C) APS Division of Particles and Fields (DPF19), Boston, MA. *Program Committee Member; Computing, Analysis Tools, and Data Handling Track Co-Chair.*

2017 (W) Future of collider searches for Dark Matter at the LPC, Batavia, IL. *Organizing Committee Co-Chair.*

2016 (W) Beyond Leading Order Calculations on HPCs, Batavia, IL. *Local Organizing Committee Member.*

2016 (C) International Symposium on Grids and Clouds, Taipei, Taiwan. *Program Committee Member.*